

Examiner reports

Q1.

The vast majority of students were able to correctly answer (a) with answers given in a variety of forms.

(b) was quite well answered, though a significant number of students failed to identify the sequence as geometric and so were unable to make any progress.

Only a small proportion of students scored full marks on (c), often because they failed to make a clear enough statement about how the total length would be affected.

Q3.

Students almost always scored full marks for their answers to parts (a)(i) and (a)(ii) but writing the n th term of the geometric series in the required form in (a)(iii) proved to be much more challenging. Although the majority of students started correctly by writing $420 \times 0.7^{n-1}$, a large proportion of these students failed to write this in the form 600×0.7^n . The most common wrong answer, for those who attempted to re-write $420 \times 0.7^{n-1}$ in the stated form, was 294×0.7^n . In part (b)(ii) there was some confusion with the term and sigma notation which resulted in the sum being equated to zero. The most successful approach was to solve $U_k = 0$ to obtain the value of the positive integer k and then to find

the value of $\frac{k}{2} (240 + 0)$, which represents $\sum_{n=1}^k u_n$, the sum of the k terms of the arithmetic series with 1st term 240 and last term $0 (= U_k)$. In a minority of cases where the correct expression for U_k was equated to zero, an incorrect rearrangement resulted in a negative value for k . Students who made this error did not seem to realise that such a value for k was meaningless.

Q4.

This opening question, which tested the use of standard formulae for a geometric series, proved to be a welcome starter with a large majority of students scoring full marks.

Part (c) was the least well answered where some students used $n = 11$ in the relevant formula for the sum, S_n , of a geometric series.

Q5.

Most students used the given formula for the n th term of the geometric series correctly, although a few assumed, incorrectly, that the first term was 48, for which some follow through allowance was made. The common ratio and sum to infinity were usually found correctly, although some students gave a value for the common ratio that was greater than 1 without ever realising that such a value was not appropriate for use in the sum to infinity formula as stated in the formulae booklet. The notation in part (d) was, as expected, found more challenging. However, the stronger students coped well, with a pleasing number using as their method the sum from the 4th term to infinity. A common error in part (d) was to assume that the value for the sum could be found by calculating $(S_\infty - S_4)$ rather than $(S_\infty - S_3)$, although the former was awarded some partial credit if correctly calculated.

Q6.

Most of the candidates who used the expected algebraic approach successfully reached the required value for the common ratio. Very few candidates quoted incorrect formulae for the third and sixth terms. Those who assumed the value of r or just showed arithmetical working rarely gained any credit in part (a)(i). A large majority of candidates scored full marks in part (a)(ii) for finding the first term of the series. Most candidates realised that the sum of the first twenty terms of the series was required and so picked up the method mark, but errors such as ' $4 \times 3^{20} = 12^{20}$ ' in intermediate working towards the printed answer was enough to lose the accuracy mark. Part (b)(ii) was not answered well, with a significant majority failing to score even a single mark. A common error was to replace u_n by the formula for the sum of the first n terms. Those candidates who did make progress usually substituted a correct expression, $4 \times 3^{n-1}$, for u_n , and a high proportion of these candidates then took logarithms and used an appropriate logarithmic law correctly. However, a significant number of these candidates failed to score the final mark because they gave the non-integer value 32.43 for n or they rounded down to 32. Some candidates used a trial and improvement approach in preference to taking logarithms with varying degrees of success. When applying a trial and improvement approach, candidates should be aware that it is insufficient to just use one value of n .

Q7.

A very large percentage of the candidates who attempted this question found the correct value for the sum to infinity in part (a).

In part (b) the majority of candidates got the correct 6th term but a large proportion felt that evaluating using a calculator (rather than changing to powers of 2 and 3) was an appropriate

way of showing that $12\left(\frac{3}{8}\right)^5$ and $\frac{3^6}{2^{13}}$ were equal. Only a very small minority of candidates confused the 6th term with the sum to 6 terms.

In part (c)(i), many candidates could state the general formula for the n^{th} term but a significant minority failed to substitute the necessary values. A number of candidates, without proof, used either their answer to part (b) or the given answer in part (c)(ii) to write down their answer in terms of powers of 2 and 3. There were relatively few fully correct solutions to part (c)(ii) though quite a few did pick up some marks.

Starting with their u_n from part (c)(i) a common error was to think that $\log ab$ was $\log a \times \log b$ rather than $\log a + \log b$ or to take the power to the front of it all. This meant that the correct use of a further log law was often the only mark that was gained. Many who correctly applied two laws of logarithms then struggled to write their expression in terms of $\log 2$ and $\log 3$ only. Although rare, there were some excellent solutions seen which differed from the candidates' normal approach.

One such approach, for full marks, is illustrated by ' $u_n = 12 \times \left(\frac{3}{8}\right)^{n-1} = 32 \times \frac{3^n}{8^n} = \frac{2^5 \times 3^n}{2^{3n}}$ ';
 $u_n = 3^n \times 2^{5-3n}$; $\log_a u_n = \log_a 3^n + \log_a 2^{5-3n} = n \log_a 3 + (5 - 3n) \log_a 2 = n \log_a 3 - (3n - 5) \log_a 2$
 Credit was given to those candidates who presented the last two stages of the above

solution but, without the first of the three stages at some point in their solution to the question, only half marks could be awarded.

Q8.

Many candidates correctly showed the printed value for r in part (a)(i) by forming and

solving the equation ' $\frac{10}{1-r} = 50$ '. Those who used verification methods rarely scored both marks. Most candidates correctly found the second term of the geometric series, but a significant number of candidates in part (b)(i) tried to 'guess' the answer for the common difference instead of using the formula for the n^{th} term of an arithmetic series and ended up with the wrong answer ' $d = 0.5$ '.

The final part of the question, which involved understanding of the sigma sign, was better answered than in previous papers but still the majority of candidates failed to obtain the

correct value. The most common wrong answer was 820, obtained by using $\frac{1}{2}n(n+1)$ with $n = 40$, presumably from scanning through the formulae booklet.

Q9.

Part (a) was normally answered correctly but those who relied entirely on a verification approach generally scored no more than 1 mark and many of these scored no marks because they failed to state a conclusion.

Most candidates were able to find the correct values for the first term and sum to infinity of the geometric series but many could make no progress in the final part of the question which involved the sigma sign. Of the better solutions, the very common error was to find the value of ' $S_{\infty} - S_6$ '. The most common correct methods involved either finding the value of ' $S_{\infty} - S_5$ ' or recognising that the sum to infinity of a geometric series with first term $u_6 (= 48.6)$ and common ratio 0.6 was required.

Q10.

A greater proportion of candidates than in previous years used the correct formulae when answering this topic, which involved an infinite geometric series. Many candidates found the correct value for the common ratio, although the incorrect value 1.25 was seen. Candidates who gave this incorrect value did not seem to be concerned that their answer for the sum to infinity was negative. In part (c) some candidates did not gain the accuracy mark because they only gave a one decimal place answer for the sum of the first twenty terms. The final part of the question, as expected, was a challenge to all. No credit was given to those who just used particular values for n . Better candidates stated that the n^{th} term was $20 \times 0.8^{n-1}$ but only the most able candidates gave the intermediate stage, $20 \times 0.8^{-1} \times 0.8n$, or equivalent, to persuade the examiners that the printed result had been obtained convincingly.

Q11.

There continues to be an improvement in candidates' general ability to answer questions on geometric series. Use of wrong formulae was less common than in previous sessions. However, far too many candidates carelessly lost negative signs within their solutions and

so failed to gain all the accuracy marks.

Q12.

The majority of candidates presented correct solutions to parts (a) and (b) although the value 36 for the common ratio appeared more often than expected. In part (c)(i), although many candidates applied the correct formula for the sum to 12 terms of the geometric sequence, others focused on the form of the printed answer and replaced n in the general formula by k . Most of these candidates failed to gain any marks for this part because they did not use $n = 12$. A very common error was ' $-4(1 - 4^{12}) = -4 + 16^{12}$ ' which resulted in the answer ' $k = 24$ '. Sign errors were also common. In general, only the most able candidates were able to see how to use the earlier parts to answer part (c)(ii) and even some of these failed to evaluate their expression to reach the correct value 67108848.

Q13.

A majority of candidates indicated that the method for solving equations of the form $a^x = b$ was well known. The most common errors are illustrated by

(a) $\log x^{0.8} = \log 0.05$

(b) (i) $x = \frac{\log 0.8}{\log 0.05}$

- (ii) answers 13.4 or 13.43 Part (b), which tested geometric series, although answered much better than the corresponding question in summer 2005, still indicates a general weakness in this area of the specification and the associated algebraic manipulation. It was not uncommon to see candidates using a specific value of a , either 1 or 20 or even 100, to solve the equation

$$\frac{a}{1-r} = 5a$$

In the final part a significant proportion of the candidature used the formula for the sum to n terms rather than the formula for the n th term. Some better candidates obtained $n = 14.5...$, but failed to realise that n was an integer.

Q14.

This question on geometric series and logarithms proved to be very difficult for many candidates, with a significant number of the weaker candidates unable to quote correct relevant formulae from the booklet. Transforming the first sentence of the question into a correct equation was beyond many, but partial credit was usually gained by using the printed value for r . This part of the question seemed to show up a lack of understanding of algebraic manipulation. Rearranging ' $4a(1 - r) = a$ ' to ' $r = 3/4$ ' proved to be more difficult than anticipated. In part (b) it was very evident that a significant number of candidates did not understand the meaning of the letter ' a ' in the formula for the sum to n terms of a geometric series. To their credit, some overcame this difficulty by working out S_1 as

$$\frac{a(1-r^1)}{(1-r)} = a \text{ and then stating 'so first term is } a'$$

Those who used the correct formula from the booklet normally obtained the method mark

but a significant proportion of these candidates could not evaluate the expression correctly on their calculators. Those candidates who resorted to listing the 10 terms and then adding them rarely obtained the correct four decimal place answer and clearly used up valuable time in the process.

Although some excellent solutions to all parts of (c) were seen, relatively a few candidates went beyond part (c)(i). Many were able to quote the correct expression for u_n from the formulae booklet. For u_{2n} a common incorrect answer was $2ar^{n-1}$. It was very common to find candidates using ar^{n-1} to mean $(ar)^{n-1}$ which, although not penalised in part (c)(i) lead to mistakes by candidates in reaching the printed answer in part (c)(ii).

Those candidates who recognised the left-hand side of part (c)(ii) as $\log_{10} \left(\frac{2a^n}{2a^{2n}} \right)$ and then cancelled the a 's were far more successful than those who did not use the law of logarithms before substituting. It was pleasing to see some candidates, having abandoned their solution to part (c)(ii), using the printed answer with $n = 100$ to gain credit in the final part of the question. For full credit, examiners expected to see some evaluation of $100 \log_{10} \left(\frac{4}{3} \right)$ before writing the printed answer, for example ' $\approx 12.49 \approx 12.5$ ' was acceptable.